

As part of the B.Tech curriculum under MG University, we, the 1st year Electrical and Electronics Engineering students of Adi Shankara Institute of Engineering & Technology, Kalady have visited the Pallivasal Hydroelectric Power Plant on 23rd February 2013. The IV was successfully completed under the able guidance of Arathy miss, Sijo sir. This training gave us a rare opportunity to understand the power distribution side and to know about the devices employed for that purpose. It is located on the Mudirapuzha River of Periyar basin in Munnar / Pallivasal village in Devikulam Taluk of Idukki district, in southern Indian state of Kerala.

The run of the river scheme envisages the efficient utilisation of available water at R.A.Head works by diverting the water through a new water conductor system for power generation in the existing 6X37.5 MW Pallivasal Power House and proposed extension power house to be located near the existing Pallivasal Power house for generation of 153.90 Mu.per annum.

The Pallivasal hydroelectric project commissioned in 1939 utilising the waters of Mudirampuzha river, the important tributary of Periyar is the first power station of the State with an installation capacity of 13500 K.W. The Sengulam hydroelectric project is situated about 9 K.M downstream of Pallivasal Project in Mudirampuzha River. 21.25 M.W Power is generated from this project. The Neriamangalam and Periyar hydroelectric Projects are the other projects.

Pallivasal electric project was commissioned in 1940. Kerala electricity transmission system also emerged in 1940 with 66 kv line from Pallivasal to Thiruvananthapuram and eight substations were also commissioned in 1940.

Pallivasal is located 8 km from Munnar in Idukki district, Kerala. This is the site of the first Hydro-Electric Project in Kerala and a place of immense scenic beauty.

<u>Attribute</u>	<u>Value</u>
Hydro Electric Project	Pallivasal Hydroelectric Project
State Name	Kerala
District	Idukki
River	Muthirapuzha/Periyar

Basin Name	West flowing rivers from Tadri to Kanyakumari
Hydro Electric Region	Southern HE Region
Total Installed Capacity (MW)	37
Type of Project	Major
Hydroelectric Project Status	Completed
Project Owner Type	State
Owner Name	KSEB
Inter Basin Project	No
Project Sharing	None
Inter State Agreement (Ratio like 50:50)	-
International Sharing	None

CLIENT:

Essar Projects Limited

Formerly, Essar Constructions (India) Ltd.

Execution: since June 2007

MAIN DATA:**Intake**

- Type : Concrete lined Trapezoidal
- Length: 115.1m
- Bottom width: 5m
- Design discharge: 22.66 cumecs

Power Tunnel

- Shape & Size : Circular & 3.50m (finished)
- Penstock No. 1 (for extension scheme)
- Diameter : 2.0 m
- Maximum discharge: 13.93 cumecs

Powerhouse

- Type: Surface Power House
- Installed capacity Stage I: 2X30 MW
- Maximum Discharge : 8.73 cumecs
- Maximum Gross head : 592.50
- Net Head : 594.92
- Type of machine : Vertical Pelton generating unit
- Annual energy output Stage I and II 4182 GWh.

Electricity production

MWh ◆	year ◆
165,170.00	2000
240,492.00	2007
422,593.00	2020

FACTS ABOUT PALLIVASAL HYDRO ELECTRIC POWER PLANT

Annual Carbonemissions2000 kg	0 kg (0 tonne) + 
Annual Carbonemissions kg	0 kg (0 tonne) + 
Annual Carbonemissionsnextdecade kg	0 kg (0 tonne) + 
Annual Energyoutput2000 MWh	165,170 MWh + 
Annual Energyoutput MWh	240,492 MWh + 
Annual Energyoutputnextdecade MWh	422,593 MWh + 
Carmald	33439 + 
City	Idukki Dist + 
Continent	Asia + 
Country	India + 
Fuel type	Hydro + 
Generation capacity electrical MW	37.5 MW + 
Intensity2000 kg CO2 per MWh elec	0 kg (0 tonne) + 
Intensity kg CO2 per MWh elec	0 kg (0 tonne) + 
Intensitynextdecade kg CO2 per MWh elec	0 kg (0 tonne) + 
Latitude	9.85 + 
Longitude	76.967 + 
Ownercompany	Kerala State Electricity Board + 
Point	9.85 N, 76.9666667 E + 
Primary fuel type	Hydro + 
State	Kerala + 
Status	Operational + 